

so that every 2023-ring with $3 = 0$ is a 3-ring and hence commutative. Finally, in $\mathbb{F}_7[T]$ we find

$$\gcd(T^{2023} - T, (T+1)^{2023} - (T+1)) = T^7 - T \in \mathbb{F}_7[T],$$

so that every 2023-ring with $7 = 0$ is a 7-ring and hence commutative by Theorem 5.5.

6. Commutativity of p^2 -rings

Morita [Mo78] has shown by direct and rather long calculations that p^2 -rings are commutative for $p \in \{2, 3, 5\}$. He also mentioned that the same method (but with much more effort) works for $p = 7$. In this section, we will generalize Morita's approach to show the commutativity of p^2 -rings with $p = 0$. The restriction to $p = 0$ (and the fact that p will be arbitrary) allows for a considerable simplification of the calculations.

Lemma 6.1. *In a p^k -ring with $p = 0$ there is an equational proof that for every element x the element $e := x + x^p + \dots + x^{p^{k-1}}$ is central.*

Proof. Observe that $e^p = e$. Hence, Theorem 5.5 tells us that e is an $\mathbb{F}_p$ -linear combination of idempotent and hence central elements. $\square$

Lemma 6.2. *Let V be an $\mathbb{F}_p$ -vector space, $U \subseteq V$ be a subspace, and $x_0, \dots, x_{p-1} \in V$ such that $\sum_{i=0}^{p-1} \lambda^i x_i \in U$ for all $\lambda \in \mathbb{F}_p$. Then $x_0, \dots, x_{p-1} \in U$.*

Proof. Consider the Vandermonde matrix $(\lambda^i)_{\lambda \in \mathbb{F}_p, 0 \leq i < p}$ over $\mathbb{F}_p$. It has an explicit inverse matrix $(u_{i,\lambda})_{0 \leq i < p, \lambda \in \mathbb{F}_p}$, so that

$$\sum_{\lambda \in \mathbb{F}_p} u_{i,\lambda} \lambda^j = \delta_{i,j}.$$

Hence, for all $0 \leq j < p$ we have

$$U \ni \sum_{\lambda \in \mathbb{F}_p} u_{j,\lambda} \sum_{i=0}^{p-1} \lambda^i x_i = \sum_{i=0}^{p-1} \left(\sum_{\lambda \in \mathbb{F}_p} u_{j,\lambda} \lambda^i \right) x_i = \sum_{i=0}^{p-1} \delta_{j,i} x_i = x_j. \quad \square$$

Definition 6.3. Let A be a ring, $x, y \in A$ and $i, j \in \mathbb{N}$. Following [FM46], $[x^i y^j]$ denotes the sum of all monomials in x, y in which x appears i times and y appears j times. There are $\binom{i+j}{i}$ of these monomials. For example, $[xy^2]$ equals $xy^2 + yxy + xy^2$. Notice that

$$(x+y)^n = \sum_{i=0}^n [x^i y^{n-i}] \quad (6)$$

for all $n \in \mathbb{N}$. Of course, if A is commutative, then $[x^i y^j] = \binom{i+j}{i} x^i y^j$, and eq. (6) is just the binomial theorem.

Theorem 6.4. *There is an equational proof that every p^2 -ring with $p = 0$ is commutative.*

Proof. In the proof, all variables are quantified over all elements of the ring. By Lemma 6.1, the element $f(x) := x + x^p$ is central. Hence, $f(x+y) - f(x) - f(y)$ is central as well. By eq. (6) this simplifies to $[xy^{p-1}] + \dots + [x^{p-1}y]$. Substituting x with λx for $\lambda \in \mathbb{F}_p$ shows that

$$\lambda [xy^{p-1}] + \dots + \lambda^{p-1} [x^{p-1}y]$$

is central. By applying Lemma 6.2 to the subspace $Z(A) \subseteq A$, all $[xy^{p-1}], \dots, [x^{p-1}y]$ are central. In an equational proof, we cannot write $Z(A) \subseteq A$ or speak of subspaces, but it is clear that the proof of the Lemma still applies: we just have to replace the formula $a \in Z(A)$ by $ab = ba$ for a variable b . In particular, $[xy^{p-1}]$ commutes with y , meaning

$$y(xy^{p-1} + yxy^{p-2} + \dots + y^{p-2}xy + y^{p-1}x) = (xy^{p-1} + yxy^{p-2} + \dots + y^{p-2}xy + y^{p-1}x)y.$$

The i th summand on the left side is equal to the $(i + 1)$ th summand on the right side. Hence,

$$y^p x = x y^p,$$

showing that y^p is central. Since $y + y^p$ is central, y must be central. $\square$

Example 6.5. There is an equational proof that every 73-ring is commutative: The set of $p \in \mathbb{P}$ with $p - 1 \mid 73 - 1$ is 2, 3, 5, 7, 13, 19, 37, 73, so by Theorem 3.14 it suffices to prove commutativity modulo these primes. We use Theorem 4.1. The exponent k is 1 for the primes 7, 13, 19, 37, 73 and 2 for the primes 2, 3, 5. So the claim follows from Theorem 5.5 for the first set of primes and from Theorem 6.4 for the second set of primes.

7. The general case

Next, we aim for the general case. Our methods for p -rings and p^2 -rings do not generalize to p^3 -rings, so that different ideas are required. The (non-equational) proofs of Jacobson's Theorem in [Wa71, NT74] derive it from Wedderburn's Theorem, which states that every finite division ring is commutative.

More precisely, they are structured as follows:

- (1) Reduce the general case to the case of finite rings.
- (2) Reduce the finite case to the case of division rings.
- (3) For division rings, cite (or prove again) Wedderburn's Theorem.

We will follow a similar approach here, but the proofs need to be made constructive (thus, avoiding LEM) and then transformed into equational logic. The common proofs of the Wedderburn Theorem such as the one by Witt [Wi31] are not constructive, though. They start with a finite non-commutative division ring of minimal cardinality and derive a contradiction. To solve this, we need to formulate the Wedderburn Theorem in a different way.

But first, let us observe the following interesting relationship:

Proposition 7.1. *In classical logic, the following are equivalent:*

- (1) *Wedderburn's Theorem is true.*
- (2) *Finite reduced rings are commutative.*
- (3) *For every $n > 1$, finite n -rings are commutative.*

Proof. (1) $\Rightarrow$ (2): It suffices to prove that every finite reduced ring A is a product of division rings. If A is trivial or a division ring, we are done. Otherwise (here we are using LEM) we can choose a non-unit $x \neq 0$. Since A is finite, there are $n, m > 0$ with $x^n = x^{n+m}$. Then $e := x^m$ is idempotent $\neq 0, 1$ and central by Lemma 2.8. Thus, $A \cong eA \times (1 - e)A$ and induction on the cardinality of A proves the claim. (2) $\Rightarrow$ (3): This is clear since n -rings are reduced. (3) $\Rightarrow$ (1): If D is a finite division ring, say with n elements, then D is an n -ring because of Lagrange's Theorem applied to $D^\times$. $\square$

The idea is now to formulate a constructive version of the Wedderburn Theorem for very special kinds of finite n -rings. We already know that it is enough to consider p^k -rings with $p = 0$ (Corollary 4.2).

Definition 7.2. Let $p \in \mathbb{P}$, $k \geq 1$, and $f \in \mathbb{F}_p[T]$ be a polynomial. The *constructive Wedderburn Theorem* $W_{p,k,f}$ is the following statement: Let A be a p^k -ring with $p = 0$ and $a, b \in A$ two elements with

$$ba = f(a)b. \tag{7}$$

Then

$$ba = ab.$$

This theorem will be an intermediate step in proving commutativity of A . It is more easy than just showing $ba = ab$ directly because of the additional assumption in eq. (7), which roughly says that a and b “almost” commute. The statement $W_{p,k,T}$ is trivial.

Lemma 7.3. *The Wedderburn Theorem implies the constructive Wedderburn Theorem $W_{p,k,f}$ for all p, k, f . In particular, $W_{p,k,f}$ holds in classical logic.*

Proof. With the notation from Definition 7.2, consider the subalgebra $\mathbb{F}_p\langle a, b \rangle \subseteq A$ generated by a, b . Define $E := \{a^i b^j : 0 \leq i, j < p^k\}$ and let M be the $\mathbb{F}_p$ -submodule of $\mathbb{F}_p\langle a, b \rangle$ generated by E . Then $a^{p^k} = a$, $b^{p^k} = b$ and eq. (7) imply that $a \cdot E$ and $b \cdot E$ are contained in M . This implies $E \cdot E \subseteq M$, so that $M \cdot M \subseteq M$ and therefore $M = \mathbb{F}_p\langle a, b \rangle$. Hence, $\mathbb{F}_p\langle a, b \rangle$ is a finite p^k -ring. So the claim follows from Proposition 7.1. $\square$

Trivially, when all p^k -rings with $p = 0$ are commutative, $W_{p,k,f}$ holds. We will prove next that the converse also holds. The proof is an adaptation of the proofs of Jacobson's Theorem in [Wa71, NT74]. The difference is that we will not use a proof by contradiction, and indeed the proof is (almost) constructive. We will use it to find an equational proof afterwards.

Theorem 7.4. *Let $p \in \mathbb{P}$ and $k \geq 1$. Assume that the constructive Wedderburn Theorem $W_{p,k,f}$ holds for all $f \in \mathbb{F}_p[T]$ of degree $< k$. Then every p^k -ring with $p = 0$ is commutative.*

Proof. Let A be a p^k -ring with $p = 0$, and let $a \in A$. Then $\mathbb{F}_p[a]$ is a finite reduced commutative ring, hence a finite product of finite fields $\mathbb{F}_{p^{m_i}}$ by the proof of Proposition 7.1. Hence, there are idempotent elements $e_i \in \mathbb{F}_p[a]$ such that $\sum_i e_i = 1$, $e_i e_j = \delta_{i,j}$ and such that the $\mathbb{F}_p$ -algebra $e_i \cdot \mathbb{F}_p[a]$ with multiplicative identity e_i is isomorphic to $\mathbb{F}_{p^{m_i}}$. They are p^k -rings, so that $m_i \mid k$. Notice that $e_i \cdot \mathbb{F}_p[a]$ is equal to the $\mathbb{F}_p$ -subalgebra of $e_i \cdot A$ generated by $e_i \cdot a$. Since $a = \sum_i e_i \cdot a$, it suffices to prove that every $e_i \cdot a$ is central in $e_i \cdot A$.

Therefore, we may assume that $\mathbb{F}_p[a]$ is a finite field, say $\mathbb{F}_{p^m}$ with $m \mid k$. Consider the $\mathbb{F}_p$ -linear map

$$d : A \rightarrow A, \quad x \mapsto ax - xa.$$

We need to show $d = 0$, because then a is central. One proves by induction on $n \in \mathbb{N}$ that

$$d^n(x) = \sum_{i=0}^n (-1)^i \binom{n}{i} \cdot a^{n-i} x a^i.$$

Apply this to $n = p^m$ and use $\binom{p^m}{i} \equiv 0 \pmod p$ for $0 < i < p^m$ and $a^{p^m} = a$. Then we get

$$d^{p^m} = d. \tag{8}$$

Next, consider for every $u \in \mathbb{F}_p[a]$ the map

$$\lambda_u : A \rightarrow A, \quad x \mapsto ux.$$

It is $\mathbb{F}_p$ -linear and commutes with d (since u commutes with a). In $\mathbb{F}_{p^m}[T]$ we have the well-known equation

$$T^{p^m} - T = \prod_{u \in \mathbb{F}_{p^m}} (T - u).$$

Hence, the same equation holds in $\mathbb{F}_p[a][T]$:

$$T^{p^m} - T = \prod_{u \in \mathbb{F}_p[a]} (T - u). \tag{9}$$

Consider the $\mathbb{F}_p$ -algebra homomorphism

$$\alpha : \mathbb{F}_p[a][T] \rightarrow \text{End}_{\mathbb{F}_p}(A), \quad u \mapsto \lambda_u, \quad T \mapsto d.$$